Coding Foundations Passport

Use the same planning record as you move from spoken directions to blocks, games, text code, and robots.

Name: ________________ **Class:** ________________ **Date started:** ________________

How this passport works

- Complete only the checkpoint your teacher assigns today; keep the same passport for the whole coding sequence.
- Pseudocode is a precise human-readable plan. It is not tied to one programming language.
- Use short commands, indentation, meaningful variable names, and repeated steps that another person can follow literally.
- A screenshot proves that code ran. Your planning, test evidence, and explanation prove what you understand.

Checkpoint 1 - Decompose and plan before blocks

Current context: Code.org route or teacher-provided task

Goal: What must the finished program make happen?

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Inputs and outputs: What information or action enters the system? What should the system produce?

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Subproblems: Break the goal into at least three smaller jobs.

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First pseudocode draft

```
START
  [first precise command]
  [next command]
  [repeat or decision if needed]
END
```

Partner literal test: Have a partner follow only what is written. Where did the plan become unclear or incomplete?

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Checkpoint 2 - Find patterns, variables, and abstraction

Repeated pattern: Which steps repeat? Describe the smallest useful repeated group.

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Iteration benefit: Why is a loop better here than copying the same commands?

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Generalize: Replace a task-specific detail with a variable or procedure so the plan can solve a similar problem.

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Variable name	Data type	Starting value	Operation or change
	number / string / Boolean		
	number / string / Boolean		
	number / string / Boolean		

Generalized pseudocode

```
SET [meaningful variable] TO [starting value]
REPEAT [count or condition]
  RUN [reusable procedure]
END REPEAT
```

Checkpoint 3 - Predict, test, and improve

Prediction	Observed result	One change	Result after change

Pseudocode revision: Copy the exact line you changed and write the improved version.

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Improvement claim: The revision improved the algorithm because...

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Before submitting this checkpoint

- I made a prediction before I ran the program.
- I changed only one instruction or value during each test.
- I revised the matching pseudocode, not only the executable code.
- My evidence names a cause, a result, and an improvement.

Checkpoint 4 - Plan a game feature, then read text code

Feature goal: Name one challenge, reward, rule, identity choice, feedback cue, or progression feature.

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Trigger and result: WHEN __________ happens, the program SHOULD __________.

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Feature pseudocode: Write the feature logic before changing the MakeCode project.

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Text-code bridge: Emergency Supply Grid

1. Open a new MakeCode Arcade project, name it Emergency Supply Grid, and select JavaScript.
2. Trace the outer loop as rows and the inner loop as columns. One full inner loop completes one row.
3. Use a string, numbers, and a Boolean. Perform addition, multiplication, comparison, and Boolean operations.
4. Change the grid dimensions, predict the total, run the program, and record evidence.

Code element	What it stores or controls	Predicted value	Observed evidence
missionName			
rows / columns			
priorityMode			
suppliesPlaced			

Checkpoint 5 - Collaborate, execute literally, and revise for RVR

Role	Student	Responsibility	Evidence due
Planner			
Programmer			
Tester / evidence lead			

Timeline

Milestone	Expected time	Done?	Adjustment
Plan approved			
First run			
Revision run			

Robot pseudocode: Write the movement plan with headings, speed, duration/distance, waits, and repeated actions.

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Literal execution result: What did the robot or peer do when the plan was followed exactly?

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Final revision: What changed, what evidence justified it, and what improved?

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Final reflection

How did pseudocode help you move between a block program, a text program, and a robot program? Use one specific example.

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